

Appendix D

Common Language Infrastructure

The *Common Language Infrastructure (CLI)*, not to be confused with *Command Line Interface* with the same acronym, is a technical standard developed by Microsoft [4, 3]. The standard specifies a language, its format, and a runtime environment that can execute the code. The main feature is that it provides a common interface between many languages and many platforms, such that programs can collaborate in a language-agnostic manner and can be executed on different platforms without having to be recompiled. Main features of the standard are:

Common Type System (CTS) which defines a common set of types that can be used across different languages as if it were their own.

Metadata which defines a common method for referencing programming structures such as values and functions in a language-independent manner.

Common Intermediate Language (CIL) which is a platform-independent, stack-based, object-oriented assembly language that can be executed by the Virtual Execution System.

Virtual Execution System (VES) which is a platform dependent, virtual machine, which combines the above into code that can be executed at runtime. Microsoft's implementation of VES is called *Common Language Runtime (CLR)* and uses *just-in-time* compilation. In this book, we have been using the `mono` command.

The process of running an F# program is shown in Figure D.1. First the F# code is compiled or interpreted to CIL. This code possibly combined with other CIL code is then converted to a machine-readable code, and the result is then executed on the platform.

CLI defines a *module* as a single file containing executable code by VES. Hence, CLI's notion of a module is somewhat related to F#'s notion of module, but the two should not be confused. A collection of modules, a *manifest*, and possibly other resources, which jointly define a complete program is called an *assembly*. The manifest is the description of which files are included in the assembly together with its version, name, security information, and other bookkeeping information.

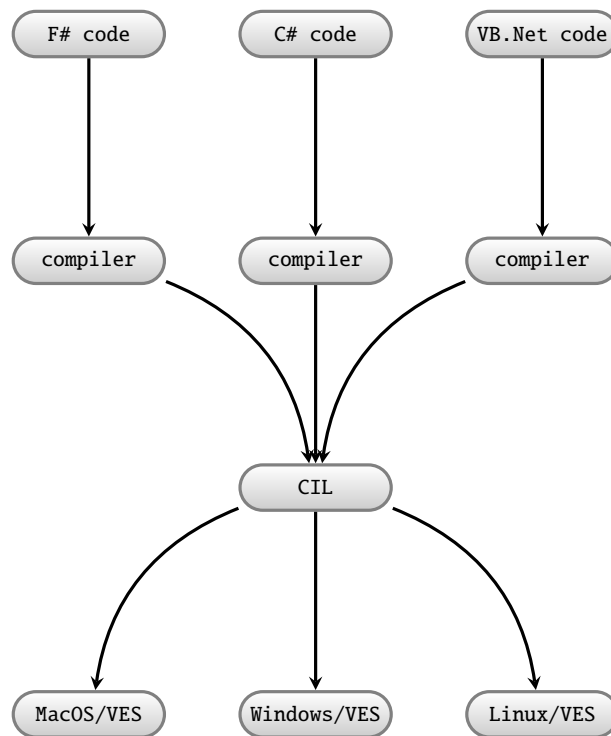


Fig. D.1 The relation between some .NET/Mono languages with the Common intermediate language (CIL), and the Virtual execution systems (VES) on some operating system (mono).